

Dohyun (Eric) Kim

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PROFILE

Penultimate Computer Systems Engineering (Hons) student at the University of Auckland with a Mechatronics adjacent focus on embedded systems and digital hardware design. Hands on project experience across PCB design, FPGA development, and embedded robotics, with strong analytical and problem solving skills.

EDUCATION

University of Auckland – Auckland CBD, New Zealand *Expected Graduation Nov 2027*

- Degree: *Bachelor of Engineering (Honours) in Computer Systems Engineering*
- Concentration: Embedded Systems & Digital Hardware Design

EXPERIENCES

Research Assistant **Auckland, New Zealand**
University of Auckland *Jul 2026 - Present*

- Develop and test **navigation and control algorithms** for **TurtleBot2** robots in a UoA robotics lab (CARES), including sensor integration and performance evaluation.
- Assist with **experiment design, data collection**, and analysis to support ongoing robotics research.

Robotics Instructor **Auckland, New Zealand**
Creative Imaginary Lab (ciLab) *Apr 2026 - Present*

- Instructed students in robot **hardware assembly and software programming** to complete assigned missions.
- Supported student development and coached teams in preparation for **nationwide robotics competitions**.

PROJECTS

Smart Energy Monitor **Embedded C**
Embedded Energy Monitoring Systems - Team Project *Oct 2025*

- Designed an embedded energy monitoring system using **ATmega328P** microcontrollers with ADC processing to measure and display real-time energy usage.
- Produced a validated PCB schematic using **Altium Designer** to achieve accurate signal conditioning.

Flappy Universe **VHDL**
FPGA Game Implementation - Team Project *Jun 2026*

- Implemented a Flappy Bird-style game in VHDL on an **Altera FPGA** with VGA output and sprite rendering to demonstrate real-time hardware design.
- Developed a pixel-priority **VHDL pipeline** to render game scenes at VGA resolution.

AWARD

3rd Place in ECSE Design Competition 2025 **Team Competition**
“Winnie the Bot” - Interactive companion robot *Sep 2025*

- Built an AI-powered robot using dual **ATmega328P** microcontrollers, an AI camera, and audio peripherals to enable face tracking, arm movement, and voice dialogue in a compact form factor.
- Contributed to **prototyping** the enclosure using iterative design to deliver a functional physical build.

LEADERSHIPS

Academic Team Executive **Auckland, New Zealand**
KEB (Korean Engineering Body) - Founding Member *Jul 2024 - Present*

- Developed and delivered **tutorial sessions** for junior engineering students across various courses.
- Participated in the planning and delivery of academic events **to support the student community**.

Official Competition Staff **NZ Robotics Olympiad 2026**
IEEE R&A (IEEE Robotics and Automation Society) *Jul 2026*

- **Selected as official Competition Staff** for NZRO 2026; ran operations and **troubleshoot technical issues**.

Full-time Student Volunteer **NZ Robotics Olympiad 2025**
IEEE (Institute of Electrical and Electronics Engineer) *Jul 2025*

- **Volunteered 40+ hours** supporting operations and logistics at NZRO 2025, an IEEE robotics competition.

SKILLS

Languages & Frameworks: C, VHDL, Python, MATLAB, Java, R, React.js

Tools & Platforms: Altium Designer, LTSpice, ModelSim, Intel Quartus Prime, Proteus, Atmel AVR, Git, AutoCAD, AWS